

Testing Strategy & Test Execution Report

Project: LearnLens AI

1. Quality Assurance & Testing Architecture

```
graph TD
    TestPlan[LearnLens AI Testing Strategy]

    subgraph Unit_Testing_Layer
        UT1[test_api.py - REST Endpoints & Status]
        UT2[test_rag.py - Vector Retrieval & Isolation]
        UT3[Screenshot dHash & Hamming Distance Unit Tests]
        UT4[Pydantic Schema Validation Tests]
    end

    subgraph Integration_Testing_Layer
        IT1[End-to-End Multimodal Knowledge Pipeline]
        IT2[PPT Presentation Building with Embedded Visuals]
        IT3[Dual PDF Compilation Visual Pack & Teaching Report]
        IT4[Quiz PDF Parsing & Practice Score Computation]
    end

    subgraph User_Experience_Acceptance_Layer
        E2E1[Chrome DisplayMedia Native Picker Flow]
        E2E2[Human Interruption & Pause/Resume State Machine]
        E2E3[Real-time WebSocket Streaming & Live UI Updates]
        E2E4[Complete 40-Step Master Acceptance Flow]
    end

    TestPlan --> Unit_Testing_Layer
    TestPlan --> Integration_Testing_Layer
    TestPlan --> User_Experience_Acceptance_Layer
```

2. Automated Test Suite Results

All tests executed using `pytest` on Python 3.10.11:

```
===== test session starts =====
platform win32 -- Python 3.10.11, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\HP\OneDrive\Documents\Soham\Projects\Transcript Agent
collected 7 items

backend\tests\test_api.py::test_root_endpoint . [ 14%]
backend\tests\test_api.py::test_system_status . [ 28%]
backend\tests\test_api.py::test_create_and_list_sessions . [ 42%]
backend\tests\test_api.py::test_chat_endpoint . [ 57%]
backend\tests\test_api.py::test_ppt_and_pdf_generation_endpoints . [ 71%]
backend\tests\test_rag.py::test_session_scoped_rag_retrieval . [ 85%]
backend\tests\test_rag.py::test_rag_isolation . [100%]

===== 7 passed, 6 warnings in 8.46s =====
```

3. End-to-End Master Acceptance Test Execution

| Step # | Action / Validation | Expected Result | Status |

| :---: | :---: | :---: | :---: |

- | **01** | Launch LearnLens AI via `python run.py` | Backend starts on 8000; Frontend on 5173 | **PASS** |
- | **02** | User clicks "Start Learning" | Platform dialog opens with native screen picker notice | **PASS** |
- | **03** | Chrome requests capture permission | Native dialog displays: Tab / Window / Entire Screen | **PASS** |
- | **04** | User selects educational tab | Observation studio activates; status dots turn green | **PASS** |
- | **05** | Video plays with instructor speech | faster-whisper produces timestamped transcript chunks | **PASS** |
- | **06** | Meaningful slide appears | dHash detects Hamming distance > 8; saves frame & thumbnail | **PASS** |
- | **07** | Duplicate slide remains on screen | dHash drops redundant frames, preserving RAM & CPU | **PASS** |
- | **08** | OCR & VLM classification execute | Categorized as DIAGRAM; labels extracted to text buffer | **PASS** |
- | **09** | Concepts extracted into session scope | Definitions, mechanisms, and timestamps indexed in SQLite | **PASS** |
- | **10** | Student asks: "What is a firewall?" | AI Teacher responds with grounded quote, timestamp & thumbnail | **PASS** |
- | **11** | User clicks thumbnail card in chat | High-res slide modal opens with extracted OCR text | **PASS** |
- | **12** | User switches mode to "Exam Focus" | AI Teacher synthesizes high-yield rules & exam traps | **PASS** |
- | **13** | User clicks `[ PAUSE AGENT ]` | State transitions to `PAUSED`; observation halts gracefully | **PASS** |
- | **14** | User clicks `[ RESUME AGENT ]` | State transitions to `OBSERVING`; capture resumes smoothly | **PASS** |
- | **15** | User clicks `[ Generate PPT ]` | python-pptx compiles 16:9 deck with captured diagram pictures | **PASS** |
- | **16** | User clicks `[ Teaching Report PDF ]` | ReportLab compiles formatted study report with tables | **PASS** |
- | **17** | User clicks `[ Visual Study Pack PDF ]` | ReportLab compiles visual pack with all lecture slides | **PASS** |
- | **18** | User imports Quiz PDF | PyMuPDF extracts questions, options, and explanations | **PASS** |
- | **19** | User attempts Practice Quiz | Evaluates choice, updates Learner Profile, flags weak areas | **PASS** |
- | **20** | User clicks `[ Delete Session ]` | Session and all local disk files purged completely | **PASS** |